

Daily Review



1 What does a social impact affect?

People's lives

2 “More tourists spend money” – type?

Economic

3 A negative effect of building work?

Habitats damaged, or
waste

4 A real Scottish renewable project?

Whitelee or Pitlochry
Dam

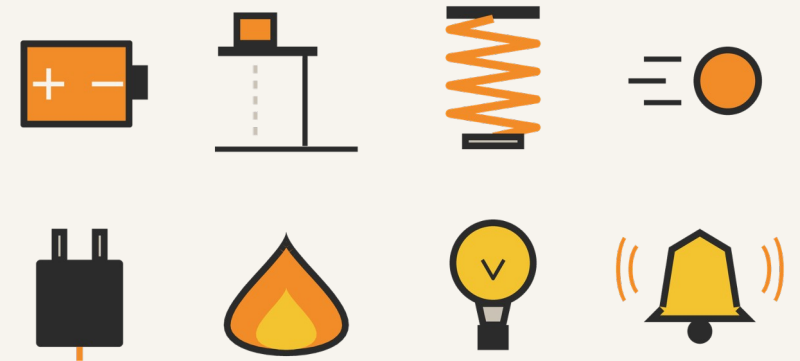
5 A strong impact answer says...?

What, why and who

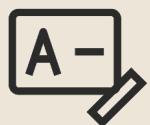
Energy In, Energy Out

Where does the energy go?

S3 Engineering Science



What the pictures mean



Show me boards
Hold up your board.



Write it
Write it in your jotter.



Turn and talk
Talk to your partner.



Hands down
No hands up. I pick.



Hands up
Hand up to answer.



Silent task
Work alone, in silence.



Jotter notes
Copy it into your jotter.



Cut and stick
Cut out and stick in.

**Whitelee's turbines don't make energy. Where does it
come from — and go?**



Whitelee Wind Farm

PART 1

Energy In, Energy Out

Name the forms, then follow the energy through a system.

By the end of this part...



Rate each one 1-3: 1 not yet, 2 nearly, 3 got it.

- 1 I can name forms of energy
- 2 I can describe energy transfers: input, useful, wasted
- 3 I can identify useful and wasted energy
- 4 I can draw a simple energy transfer diagram
- 5 I can explain my ideas clearly using what + why

SECTION A

Eight Forms of Energy

A battery, a skier, a spring — what do they have in common?

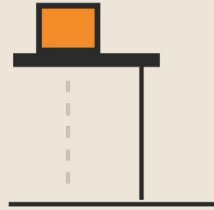
Three Energy Stores

Store: energy waiting to be used.



Chemical

in fuel, food and
batteries



**Gravitational
potential**

anything high up



Elastic

springs, a drawn
bow

Five Forms on the Move

Energy is measured in joules (J).



Kinetic

moving
things



Electrical

circuits,
the mains



Heat

makes
things
hotter



Light

lamps, the
Sun



Sound

vibrations

Worked Example – A Skier

1 Top

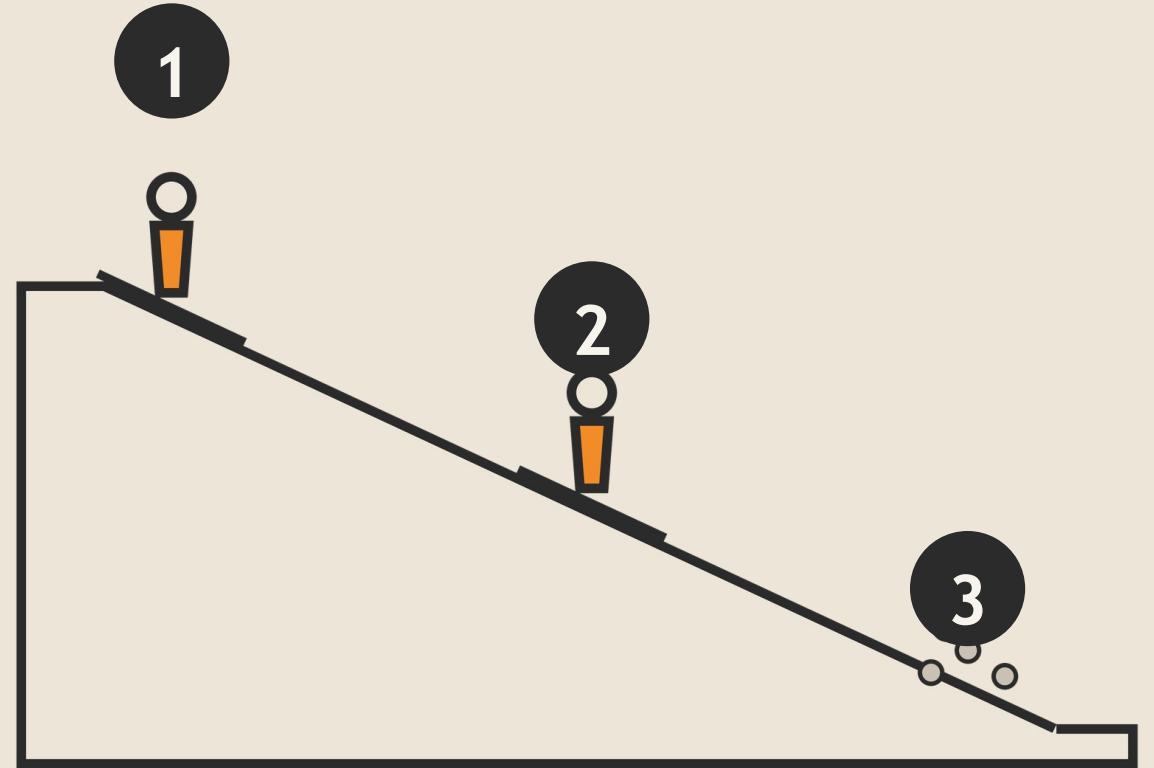
gravitational potential

2 Skiing down

kinetic

3 The snow

heat and sound – wasted



Show Me — Which Form?



- 1 A full petrol tank
- 2 A book on a high shelf
- 3 A squashed spring in a clothes peg
- 4 A spinning washing-machine drum
- 5 The unit of energy

Chemical

Gravitational potential

Elastic

Kinetic

Joules (J)

Show Me — Which Form?

EXTRA PRACTICE



- 1 A radiator warming a room
- 2 A school bell ringing
- 3 A drawn bow before the arrow flies
- 4 Two forms for a moving train
- 5 Two forms for a charging phone

Heat

Sound

Elastic

Kinetic, sound (and
heat)

Electrical, chemical

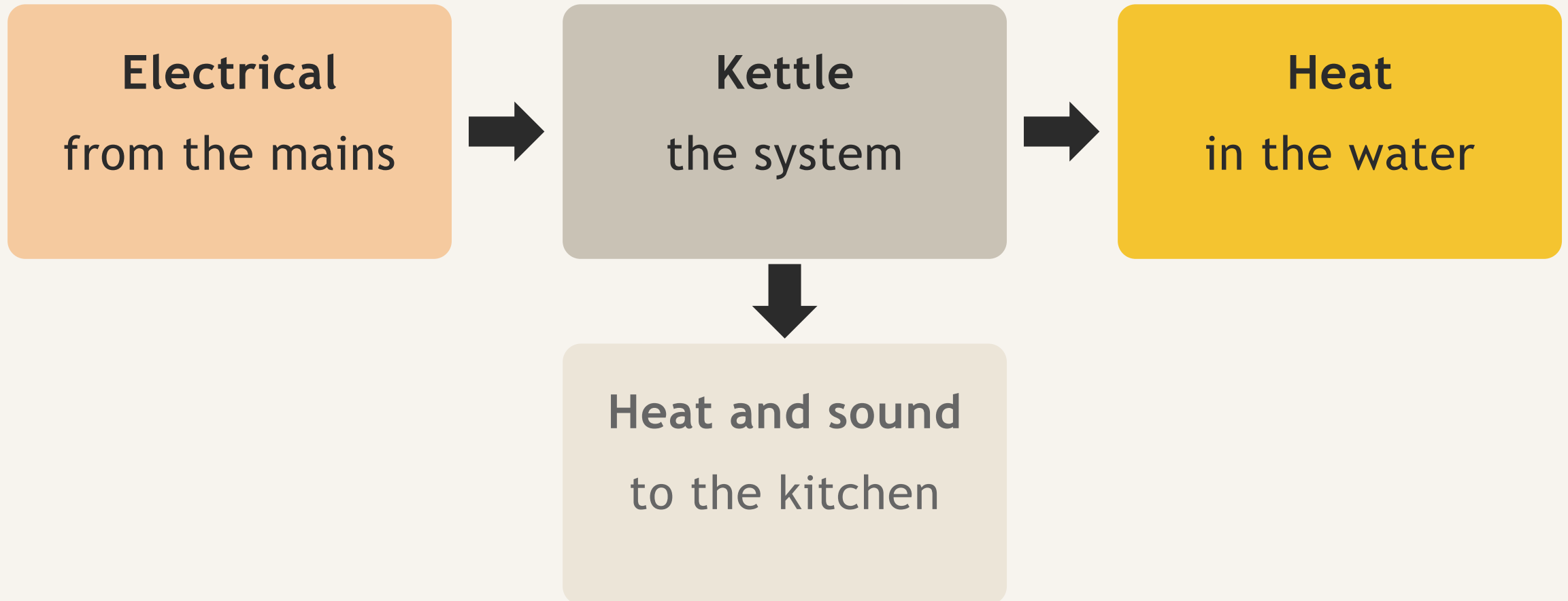
SECTION B

Where Does It Go?

Energy is never used up. So where does it go?

Input, Useful, Wasted – Electric Kettle

Energy is transferred, never used up.



Worked Example – The Input Is What Goes IN

Trap: electrical is the OUTPUT, not the input.

Input

kinetic energy of the wind

Useful output

electrical

Wasted

sound and heat



Useful Depends on the Job

Useful = it does the device's job.

Heat

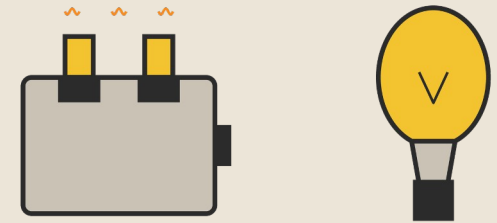
useful in a toaster · wasted in a lamp

Light

useful in a lamp · wasted in a toaster

Sound

useful in a smoke alarm · wasted in a kettle



Show Me — Input, Useful, Wasted



1 Kettle: the input energy?

Electrical

2 Lamp: the useful output?

Light

3 Lamp: one wasted form?

Heat

4 Wind turbine: the input energy?

Kinetic (of the wind)

5 What happens to wasted energy?

Spreads out as heat

Show Me — Input, Useful, Wasted

EXTRA PRACTICE



1 Petrol car: the input energy?

Chemical (in the petrol)

2 Petrol car: the useful output?

Kinetic

3 Toaster: is the light useful or wasted?

Wasted

4 Smoke alarm: is the sound useful or wasted?

Useful

5 Why does wasted energy still cost money?

You paid for the input

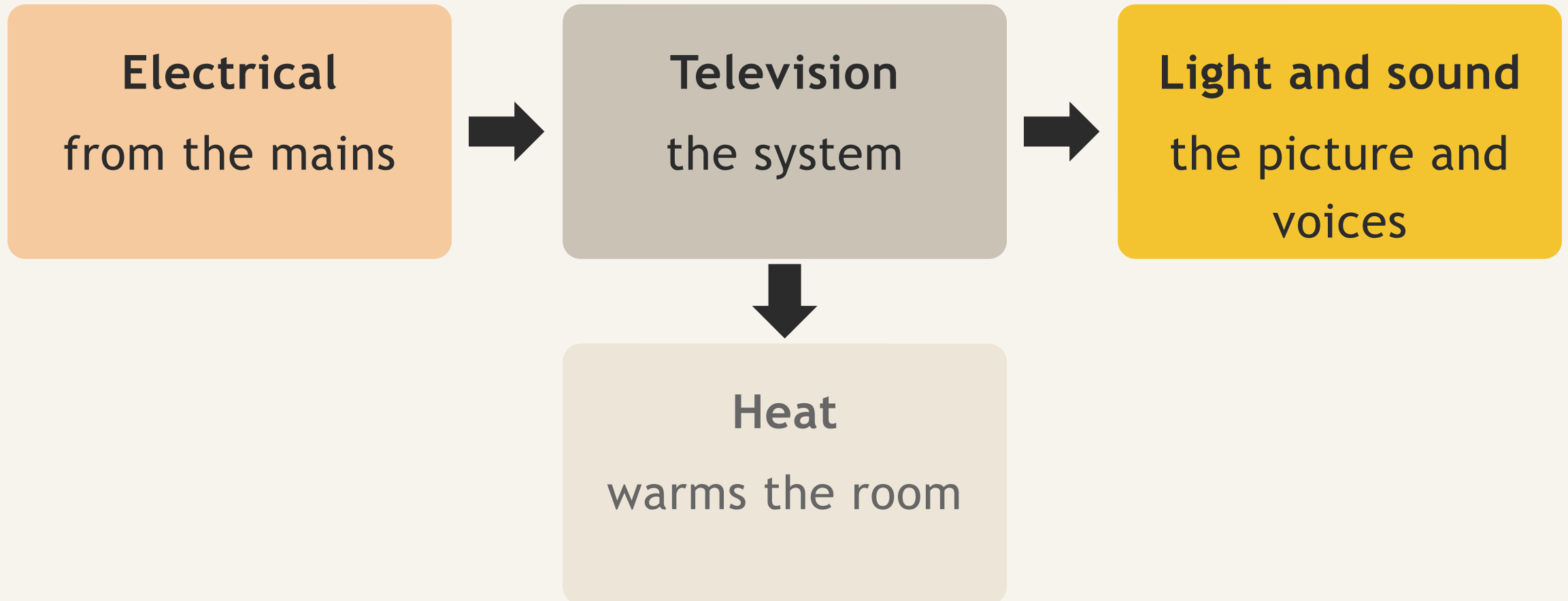
SECTION C

Draw It

Two marks for a diagram. Where does the second mark go?

Worked Example – A Television

Mark 1: input + useful. Mark 2: wasted on its own arrow.



Where the Marks Go – A Laptop

A – 2 marks

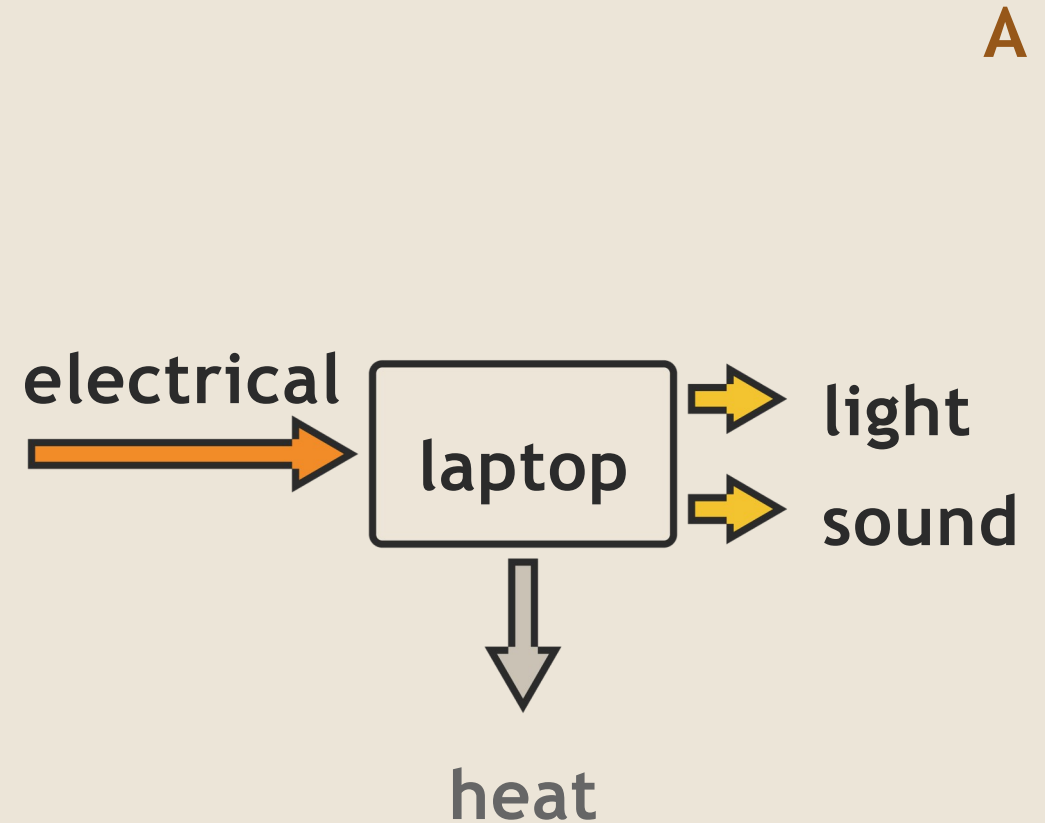
heat on its own wasted arrow

B – 1 mark

heat drawn as a useful output

C – 0 marks

only “electrical → laptop”



Show Me — Draw It on Your Board



1 Toaster

Electrical $\rightarrow$ heat; light
wasted

2 Phone charger

Electrical $\rightarrow$ chemical;
heat wasted

3 Petrol car

Chemical $\rightarrow$ kinetic;
heat, sound wasted

Show Me — Draw It on Your Board



4 Candle

Chemical → light; heat
wasted

5 How do you show the wasted energy?

Its own arrow, going
down

Show Me — Draw It on Your Board

EXTRA PRACTICE



1 Electric kettle

Electrical → heat;
sound wasted

2 Television (from memory)

Electrical → light,
sound; heat wasted

3 Wind turbine

Kinetic → electrical;
sound wasted

Show Me — Draw It on Your Board

EXTRA PRACTICE



4 Gas cooker

Chemical → heat; light
wasted

5 Why a separate wasted arrow?

It isn't the job — mark 2

What energy does a battery store?



- A electrical
- B chemical
- C kinetic
- D light

Answer B: It stores chemical energy. Electrical is what comes out.

Where does wasted energy go?



- A it is used up
- B it disappears
- C it spreads into the surroundings as heat
- D it turns back into electrical

Answer C: Energy is never used up — it spreads out, mostly as heat.

What is the INPUT energy of a wind turbine?



- A electrical
- B gravitational potential
- C sound
- D kinetic

Answer D: The wind's kinetic energy goes in. Electrical comes out.

The sound from a smoke alarm is...



- A useful
- B wasted
- C both
- D not energy

Answer A: Warning people is its job, so the sound is useful here.

Which lamp diagram gets both marks?



- A electrical → light + heat, both useful
- B electrical → light; heat on its own arrow
- C electrical → lamp
- D light → lamp → electrical

Answer B: Only B shows the heat as wasted, on its own arrow.

Mini Test 3



Mini Test 3 sheet

On your own, in silence, pen only.

10 minutes – 8 marks.

Then swap and mark from the next slides.

Mini Test 3 — Marking

ANSWERS

Q1 (1 + 1 marks)

(a) chemical — not electrical. (b) gravitational potential — not just “potential”.

Q2 (1 + 1 marks)

(a) chemical (in the petrol). (b) kinetic — not “movement”.

Mini Test 3 — Marking

ANSWERS

Q3 (1 + 1 marks)

(a) heat. (b) light or sound, because it doesn't help to brown the bread.

Q4 (1 + 1 marks)

Mark 1: electrical in, sound useful out. Mark 2: heat on its own wasted arrow.

Scores 0

a bare “light” in Q3(b); heat drawn as useful for mark 2

Brain Dump



In your jotter

Write everything you remember about energy in and energy out.
On your own, 2 minutes. No notes.
Then we check it against the Plenary.

Plenary – Energy In, Energy Out



- 1 Name the three energy stores.
- 2 The input energy of a petrol car?
- 3 Heat: useful in one device, wasted in another?

Chemical, grav.
potential, elastic

Chemical

Toaster / lamp

Plenary – Energy In, Energy Out



4 Draw a toaster's diagram.

Electrical → heat; light
wasted

5 Is energy ever used up?

No – it is transferred

Plenary – Energy In, Energy Out

EXTRA PRACTICE



1 The unit of energy?

Joules (J)

2 Form stored in a stretched spring?

Elastic

3 The two most common wasted forms?

Heat and sound

4 What does the second diagram mark need?

Wasted on its own
arrow

5 A hairdryer: the input energy?

Electrical

Look back — by the end of this part...



Rate each one 1-3 again. Compare with the start.

- 1 I can name forms of energy
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LOOKING AHEAD

Lesson 4 — Efficiency and Power

Efficiency, power, the marble run and teamwork.

Credits

The big question (Whitelee Wind Farm): ©

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Worked Example – The Input Is What Goes IN:

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